A Compact and Constant Time HQC Realization on Cortex-M4

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Abstract. In this paper we show the first constant-time implementation of HQC targeting Cortex-M4 microcontrollers. We demostrate the efficiency of our implementation and its constant-time behavior empirically, proving that HQC is a viable candidate for the PQC standardization. We then compare the performance of our implementations with other post-quantum schemes, showing the efficiency of our approach.s Key point: first Cortex-M4 optimized implementation of HQC, fully constant time

Keywords: Post-quantum cryptography, code-based cryptosystems, constant time implementation, HQC

1 Introduction

Post-quantum cryptography (PQC), which is also known as quantum-resistant cryptography, aims at defining new public-key cryptographic schemes based on mathematical problems proven to be hard against adversaries equipped with both classical and quantum computers. Recent advancements on quantum computer technology have witnessed a steep increase in scientific and engineering activities to deliver new solutions to some of the most pressing scientific and societal challenges [3]. However, from a cybersecurity standpoint, the availability of a quantum computer with a sufficiently large number of qubits, could leave most of the current protocols used to guarantee the confidentiality and authenticity of communications and storage of data vulnerable to cryptanalytic techniques based on Shor [15]-like quantum algorithms that can subvert the security guarantees of the underlying public key cryptosystems employing only a polynomial computational effort w.r.t. the exponential one required by classical computational platforms. Since 2017, several institutional and governmental bodies (e.g., The Center for Advanced China Research - CACR, The European Quantum Communications Infrastructure - EuroQCI - initiative) promoted the analysis of quantum-secure systems, but most market stakeholders are focused on tracking efforts being pursued in the PQC standardization contest promoted by the U.S. National Institute of Standard (NIST). Through two rounds of this evaluation process, NIST selected seven finalists and eight alternate candidates, which were then thinned down to two standard candidates and four alternates for a final round of cryptanalysis and possible standardization. The

current standardization-elected candidate Key Encapsulation Method (KEM) is Kyber [9] is a lattice-based cryptosystem, while NIST selected four candidates relying on different computationally hard problems, in order to obtain resilience against potentially disruptive attacks on lattice-based cryptosystems. Among the four candidates, a polynomial-time cryptanalytic attack eliminated SIKE, leaving only three KEMs which are based based on the hardness of the syndrome decoding problem, i.e., Classic McEliece [2], BIKE [1] and HQC [10]. Among these candidates, only HQC can exhibit a security proof based on a polynomial reduction of the solution of a computationally hard problem to the breaking of the cryptosystem. In particular, any successful break of HQC implies a faster way to solve the syndrome decoding problem on a random quasi-cyclic code of rate $\frac{1}{2}$. By contrast, Classic McEliece relies also on the indistinguishability of a hidden Goppa code from a random code, and BIKE relies on the indistinguishability of a hidden QC-MDPC code from a random quasi cyclic code. While both arguments are currently unchallenged, there is no proof that they are indeed corresponding to computationally hard problems.

Hamming Quasi-Cyclic (HQC) [10,11] is a code-based IND-CCA2 secure KEM whose security relies on the syndrome decoding problem for quasi cyclic codes. It is obtained by applying the HHK transform [7] on a IND-CPA construction, which is also denoted as HQC.PKE. HQC uses two types of codes: a tensor code (obtained concatenating a Reed-Muller and Reed-Solomon code) with a public generator matrix G and a random double-circulant $[2n, n]$-code with a systematic parity check matrix $(1, h)$. It is worth noting that the public code with generator matrix G is only employed for its error correcting capability, and plays no role in the security proofs (indeed, it is public).

Given the sound security record of HQC, much attention has been devoted to its side-channel cryptanalysis. Most of the independent analyses focused mainly on power analysis or on extracting electromagnetic traces [4,5,13,14], whilst by the end of the 2nd round of the U.S. NIST contest the authors in [16] and [12] presented a practical key recovery strategy based on the effectiveness of a chosen ciphertext timing attack against the non-constant time implementation of the decoder of the Bose-Chaudhuri-Hocquenghem (BCH) code (which was employed before the current concatenated Reed Mueller-Reed Solomon code). Both approaches exploit the dependence between the running time of the decoding procedure and the number of decoded errors. Specifically, when considering the HQC instance exhibiting 128-bit security parameters, the attack in [12] requires 400 Mdecryption runs, while the one in [16] reaches a key recovery after just $5,441$ calls with 93% success rate. Both attacks proposed as a solution the design of a constant-time decoding algorithm without introducing a significant penalty in terms of computational complexity. Nonetheless, in [6] the authors described a timing-based key-recovery attack against HQC based on an implementation weakness in the step that used a procedure called *rejection sampling* during randomness generation. As a countermeasure, a more careful implementation of the rejection sampling step is effective to prevent the attack.

To date, no dedicated, constant time implementation of HQC targeted to embedded systems has been proposed, despite the fact that timing attacks are easier to carry out against them. Moreover, we also note that most of the literature focused on side-channel attacks aimed at key retrieval, while it is also possible to perform a message-retrieval attack against the encryption primitive, in case the operations in the said encryption are not constant time. Indeed, the attack mechanism mimics the common attacks on modular exponentiation, as the non-constant time polynomial multiplication techniques, such as the one in the reference code by HQC, have a loop whose body contains branches depending on the values of one of the operands. Since, in the HQC.PKE encryption primitive, one of such operands is secret (as it is the plaintext to be encrypted), providing constant time multiplication algorithms is needed to secure the encryption primitive. Furthermore, as the HHK transform requires a execution of the HQC.PKE encryption function during the computation of the HQC KEM decryption, the requirement for constant time multiplication implementations is further strengthened.

The current state of the software HQC implementations tailored to the ARM Cortex-M4 microcontroller (chosen by NIST as the reference platform for microcontroller benchmarks) includes only the one in PQM4 [8] a public software project which collects the implementation of all NIST fourth round candidates tailored for the Cortex M4 CPUs. In the PQM4 project, which is currently being used by NIST as one of the third party benchmarking suites, the latest version of the HQC implementation is directly derived from the third round reference implementation, without further tailored optimizations. As a consequence, the implementation is vulnerable to the timing side-channel attack by [6].

Our Contribution. In this paper, we present the first fully constant-time software implementation of HQC, together with a comparative analysis of constant time binary polynomial multiplication techniques, tailored to Cortex-M4 microcontrollers. We provide an experimental validation of the constant time property via a battery of statistical t-tests, while varying the CPU frequency and memory latencies, to highlight the stability of our solution across different memory access latencies (which depend on the CPU frequency). Our results show that employing a FFT based multiplication algorithm is the optimal choice for constant-time multiplications, in contrast with classic divide-et-impera techniques (e.g., Karatsuba) which provide a better tradeoff for x86_64 computing platforms.

2 Background

3 Constant time, High Efficiency HQC

4 Experimental Evaluation

5 Concluding Remarks

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